

Trend Analysis Explanation for ResearchLens

This document explains how ResearchLens detects topics and determines whether each topic is rising, declining, or stable.

1. What input does the system use?

The system starts from the uploaded papers. Each paper is represented with at least:

- `id` — unique paper identifier
- `title`
- `year` — publication year
- `abstract`
- `content` or full text

These papers are the raw input for the analysis pipeline.

2. How are topics detected?

Topic detection is performed in Module 2 of the backend pipeline.

Inputs for topic detection

- `papers`: the full list of uploaded papers
- A semantic model that converts paper text into embeddings

What the module does

- It groups papers into coherent research topics using semantic similarity.
- Each topic contains:
 - `topicId` — a unique topic identifier
 - `name` — the topic label
 - `keywords` — core keywords for that topic
 - `paperIds` — the list of paper IDs assigned to that topic

So the detected topics are not hardcoded. They come from the actual papers and the relationships between them.

3. How is trend status determined?

Trend detection happens in Module 4.

Inputs for trend detection

- `papers`: the same paper list used for topic detection
- `topics`: the topic assignments created by Module 2

What the module computes

For each topic:

1. Collect all papers assigned to that topic.
2. Count how many of those papers were published each year.
3. Build a `yearlyCounts` series such as:
 - `[{ year: 2019, count: 2 }, { year: 2020, count: 5 }, ...]`
4. Compute a linear trend slope across year vs count.

The exact code uses a simple linear regression formula:

- `x` = year
- `y` = number of papers in that year

The slope is calculated using:

- `slope = (n * sum(x*y) - sum(x) * sum(y)) / (n * sum(x^2) - sum(x)^2)`

Trend label thresholds

The trend label is assigned from the slope value:

- `slope > 0.08` → `rising`
- `slope < -0.08` → `declining`
- otherwise → `stable`

That means:

- A positive upward slope stronger than `0.08` indicates a rising topic.
- A negative slope lower than `-0.08` indicates a declining topic.
- Small or flat slope values are treated as stable.

4. Why does the app say 5 Topics Detected and 1 rising?

That means:

- Module 2 produced 5 distinct topic groups from the uploaded papers.
- Module 4 analyzed the publication counts for those topics.
- Only one topic had a slope large enough to cross the rising threshold.
- The remaining topics were either stable or declining based on their year-over-year publication trend.

5. How the final trend object looks

Each topic trend result contains:

- `topicId`
- `topicName`
- `yearlyCounts` — paper counts per year
- `slope` — the computed growth slope
- `trend` — one of `rising`, `declining`, or `stable`

6. Summary

ResearchLens finds topics by grouping similar papers into topics. Then it decides rising/stable/declining by tracking how many papers appear per topic each year and computing a linear slope over time. The final label is based on whether that slope is strongly positive, strongly negative, or close to flat.

File generated

This explanation is saved in `trend-analysis-explanation.md` in the project root.